

CSCI 332, Fall 2026

In-class Activity

In this activity, you will analyze the runtime of the gradeschool algorithm for multiplying two numbers x and y , where x has m digits and y has n digits.

1. First, let's consider the grade-school algorithm for multiplication. This algorithm works by multiplying each digit of the second number by the entire first number, shifting appropriately, and then summing the results.

For example, to multiply 5281 by 23492, we would compute (you should fill this in):

$$\begin{array}{r} 5281 \\ \times 23492 \\ \hline \end{array}$$

2. Check your answer with another group. Do they agree?
3. What “milestones” would be good to count in in this algorithm to get a sense of its runtime? What is the runtime of this algorithm in terms of m and n ?
4. Is the runtime different in a best and worst case? Try to give Θ bounds for each.

5. What are m and n in terms of the values of x and y ? For example, if $x = 5281$ and $y = 23492$, what are $m = 4$ and $n = 5$. What is the general formula for m and n ?
6. In the mid-1950s, Andrei Kolmogorov, one of the giants of 20th century mathematics, publicly conjectured that there is no algorithm to multiply two n -digit numbers in subquadratic time (less than $O(n^2)$). Does your group (circle one):
- Agree: $O(n^2)$ is the best we can do.
 - Disagree: There is a better algorithm (for example, $O(n^{1.5})$ or $O(n \log n)$).
 - We still don't know: no one has been able to come up with a proof either way.